General comments on EE2-21 Feedback Systems paper 2016

Overall, the students have done well on the exam paper, scoring an exam average of 61%.

1. The students have done a little less well on this question, scoring approximately 57%.
   (a) This is an op–amp modeling and proportional controller design question and is a variation on typical study group questions.
      i. Typical study group question, tests understanding of transfer functions for simple circuits.
      ii. Typical study group question.
      iii. Typical study group question.
      iv. Design part, relatively simple since the design involves a proportional controller.

   (b) This is a Nyquist diagram analysis and design question and is somewhat typical of study group questions, though the system is unstable.
      i. Typical study group question.
      ii. Typical study group question.
      iii. Typical study group question, a bit tricky since it requires an understanding that unstable systems require a minimum value of gain for closed–loop stability.
      iv. Qualitative analysis and design question. The students need to demonstrate an understanding of the compromise between transient and steady–state performance.

2. This is a Nyquist diagram analysis and design question that involves a system with a non–minimum phase factor. The students did not do very well on this question scoring approximately 50%, possibly because of the design compromises requested.
   (a) A typical study group question, requires the students to also demonstrate an understanding of the Routh–Hurwitz stability criterion.
   (b) Standard study group question, requires the students to relate the stability margins to both robustness and transient performance.
   (c) Standard Nyquist analysis study group question. Requires an understanding of the relation between the Nyquist diagram and steady–state performance.
   (d) The question requires an analysis of the Nyquist diagram to decide which aspects of the performance require improvement (stead–state versus transient) and to decide what class of controller can achieve that.

3. This is a question about proportinal control system design using pole placement and the characteristic equation. The students did very well on this question, scoring approximately 77%.
   (a) Standard study group question.
   (b) Standard study group question.
   (c) Standard study group question, ensures understanding of the Routh–Hurwitz stability criterion.
   (d) Standard study group question.
   (e) Standard study group question.